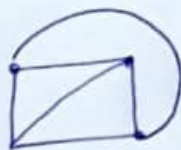


Module-4

Planar Graphs

- *1) a graph 'G' is said to be planar if there exists some geometric representation of 'G' which can be drawn on a plane such that no two of its edges intersect.
- *1) A graph that cannot be drawn on a plane without crossover b/w its edges is called nonplanar.
- *1) A graph that can be drawn on a plane without crossover b/w its edges is called planar.
- *1) drawing of a geometric representation of a graph on any surface such that no edges intersect is called embedding.
- *1) to prove a graph 'G' is non planar we have to show that all possible geometric representations of 'G' none can be embedded in a plane.

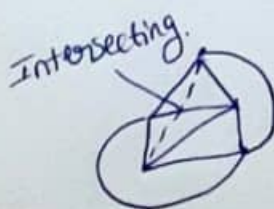
Eg:



Planar graph



Complete graph of vertex 5 i.e K_5
is non planar.



How we can tell given graph is planar or not?

$G(V, E, \varphi)$ = abstract notation

$$V = \{a, b, c, \dots\}$$

$$E = \{1, 2, 3, \dots\}$$

$$\varphi \rightarrow \begin{bmatrix} 1 & - & (a, c) \\ 2 & - & (c, d) \end{bmatrix}$$

K_5 — complete graph on 5 vertices are non planar

$K_{3,3}$ — complete graph on 6 vertices are non planar,
Bipartite graph

Kuratowski's two graphs:

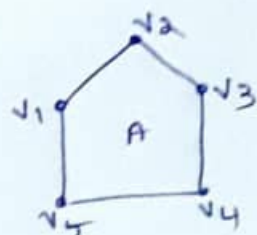
Theorem: The complete graph of '5' vertices is nonplanar.

Proof: *1 G is a complete graph with 5 vertices

$G = V_1, V_2, V_3, V_4, V_5$ — 5 vertices

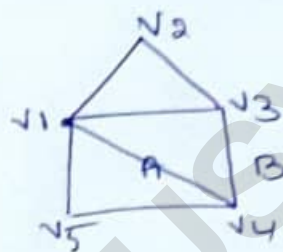
*2 let's assume we have arranged V_1, V_2, V_3, V_4, V_5 in G .

*3 we know that in complete graph every vertex is of degree $(n-1)$



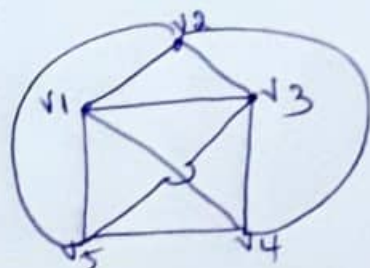
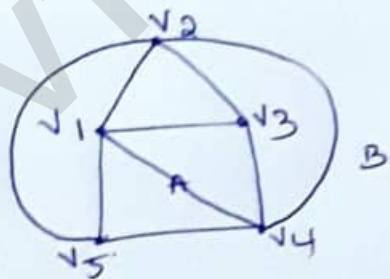
G

*4 This edge divides the region into A & B.



$n=5$ so $d(V_i)=4$.

*5 add an edge b/w V_1 & V_3
it can be in region A or B



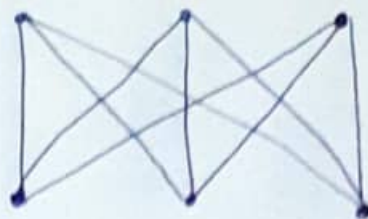
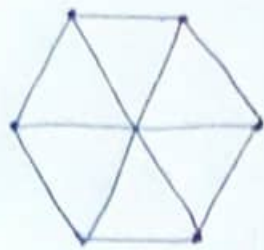
*6 we can't draw an edge b/w V_3 & V_5 without intersecting the edges on the region A or region B $\therefore$

The complete graph of five vertices is nonplanar.

Theorem 5-2.

Kuratowski's second graph ($K_{3,3}$) is also nonplanar.

  $K_{3,3}$



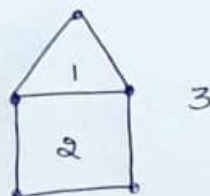
- 1] K_5 & $K_{3,3}$ are regular graph (connected graph also)
(graph in which every vertex of same degree)
- 2] both K_5 & $K_{3,3}$ are non planar
- 3] Removal of an edge or a vertex makes each of them a planar graph.
- 4] K_5 - non planar graph with smallest no of vertices
 $K_{3,3}$ - non planar graph with smallest no of edges.

(Infinite region is ^{known as} unbound, outer, exterior)

(also called as windows, faces or meshes)

Regions :- A plane representation of graph divides the plane into regions.

Fig:



*1 Region is characterized by set of edges or set of vertices.

*2 we can't defined for nonplanar graph.

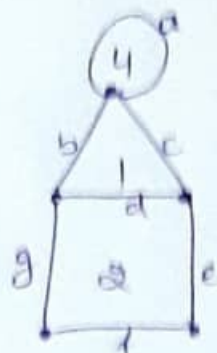
*3 can't be defined if planar graph is not embedded into a plane.

*) portion of ~~the~~ plane lying outside the graph

embedded in a plane which is infinite then it is called infinite Region.

Region continues

Eg:



*1 Region 1 is characterized by edges $\{b, c, d\}$

*1 Region 2 is characterized by edges $\{d, e, f, g\}$

*1 Region 3 is characterized by edges $\{e, f, g\}$

*1 Region 4 is characterized by edges $\{a, b, c\}$

Theorem 5-4:

A graph can be embedded in the surface of a sphere iff it can be embedded in a plane.

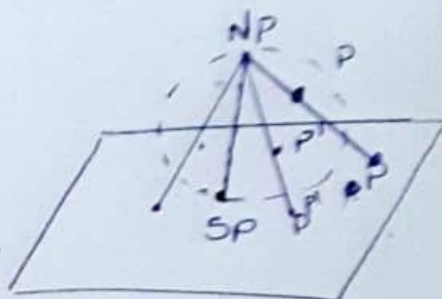
Proof: to prove this we use ^{one} stereographic projection.

1) Take the projection of the sphere on the plane

2) next take the projection of plane on the sphere.

*1 Put the sphere on the plane & call the point of contact SP

*1 at SP draw a line perpendicular to the plane



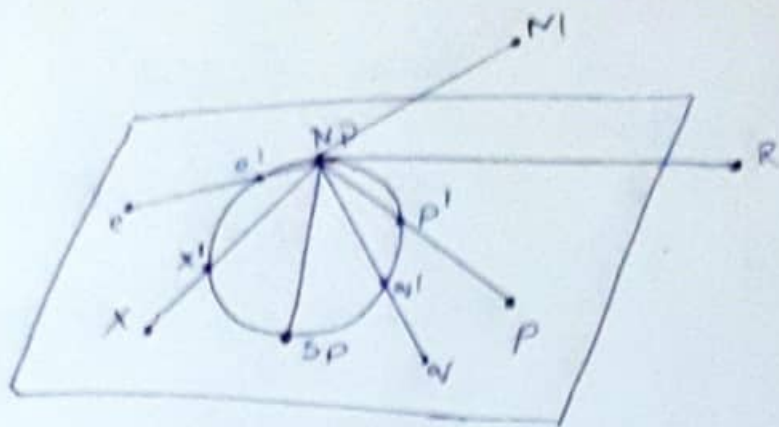
*1 Let the point where this line intersects the surface of the sphere be called NP (North Pole.)

*1 Now any point on the plane, there exists a unique point P' on the sphere and vice versa.

*1 P' is the point at which the straight line from P to point NP intersects the surface of the sphere.

*1 Thus there is one to one correspondence b/w the points of the sphere and the finite points on the plane.

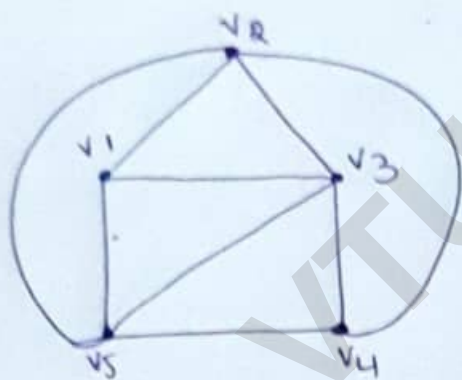
*1) Points at infinity in the plane correspond to the point NP on the sphere.



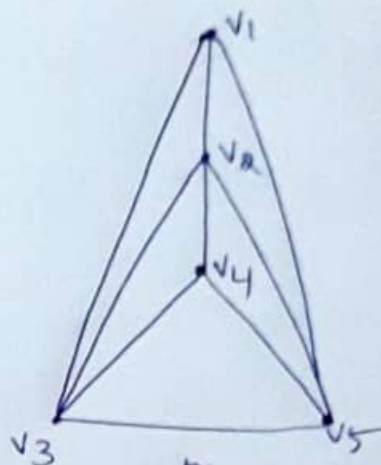
Stereographic Projection

⑥ Theorem 3-3' (Fóry theorem)

Any simple planar graph can be embedded in a plane such that every edge is drawn as a straight line segment.

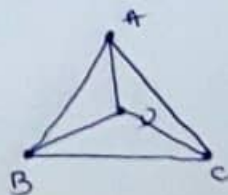
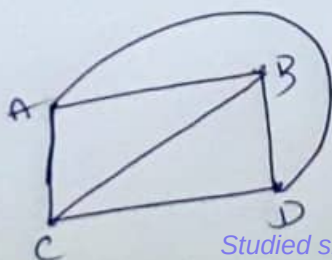


(a)



(b)
straight line representation
of the graph in (a)

*1) ^{(b) is} Simple graph a self-loop or parallel edges cannot be drawn by a straight line segment.



Theorem 5-5:-

A planar graph may be embedded in a plane such that any specified region (ie specified by the edges forming it) can be made the infinite region

- *) Thinking in terms of the regions on the sphere we see that there is no real difference b/w infinite and finite region on the plane
- *) we talk of the regions in a plane representation of a graph, we include infinite region.
- *) we also know there is no essential difference b/w an embedding of a planar graph on a plane or on a sphere.
- *) a plane is regarded as the surface of a sphere of infinitely large radius.
- *) Since a planar graph may have different plane representations, we may ask if the number of regions resulting from each embedding is the same.
- *) yes no of regions resulting from each embedding is the same. it's known as Euler's formula.

Euler's Theorem: (5-6)

A connected planar graph with n vertices and e edges has $e - n + 2$ regions.

Proof:

*1) if f is the number of regions then,

$$f = e - n + 2 \quad \text{or} \quad f + n - e = 2$$

*1) Take simple connected graph

if we have Self loop and/or Parallel edges
it always increases no of edges so we will consider simple connected graph. no pendant vertex also

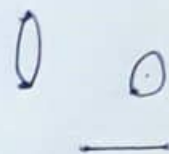
*1) Remove those edges which are not boundary of any regions

*1) Remove these 3 conditions

1) no self loop

2) no parallel edges

3) no pendant vertex.



*1) Any simple planar graph can have plane representation such that each edge is a straight line.

*1) each region can be considered as a polygon.

*1) Let f no of regions or faces

k_i no of i sided region

$$e = 3 \cdot k_3 + 4 \cdot k_4 + 5 \cdot k_5 + \dots + n \cdot k_n = 2e$$

*1) k_3 - no of triangles

k_4 - no of quadrilaterals - total

*1) each edge is boundary for two regions

$$f = k_3 + k_4 + k_5 + \dots + k_n \quad (2)$$

each polygon is a region and creating adding these region creates one more big region it's nothing but sum of triangles sum of quadrilaterals

*) There are n vertices in the graph. Sum of all the angles at each vertex is $2\pi n$ or $2n\pi$ — (3)

*) total angle at each vertex $= 2\pi$

*) Sum of all interior angles + sum of all exterior angle.

*) If we have P sided polygon

$$\text{total interior angle} = \pi(P-2)$$

$$\text{total exterior angle} = \pi(P+2)$$

$$\begin{aligned} & \text{exterior} - \text{interior} \\ & = \pi(P+2) - \pi(P-2) \\ & = \pi P + 2\pi - \pi P + 2\pi \\ & = 4\pi \end{aligned}$$

*) sum of all angles = interior angles of $f-1$ finite region + exterior angle of polygon representing infinite region

$$= \pi(3-2)k_3 + \pi(4-2)k_4 + \dots + \pi(9-2)k_9 + 4\pi$$

$$= 3\pi k_3 - 2\pi k_3 + 4\pi k_4 - 2\pi k_4 + \dots - \pi k_9 - 2\pi k_9 + 4\pi$$

$$= \pi(3k_3 + 4k_4 + \dots + k_9) - 2\pi(k_3 + k_4 + \dots + k_9) + 4\pi$$

$$= \pi(2e) - 2\pi f + 4\pi$$

$$= \pi(2e - 2f) + 4\pi \quad \text{--- (4)}$$

Equate expression (3) to (4)

$$2n\pi = \pi(2e - 2f) + 4\pi$$

$$2n\pi = 2\pi(e - f + 2)$$

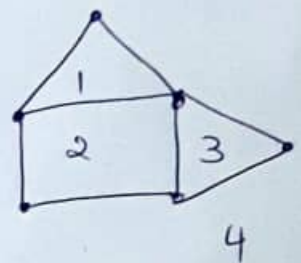
$$n = e - f + 2$$

or

$$f = e - n + 2$$

$$\boxed{f + n - e = 2}$$

Hence proved.



4

$$n = 6$$

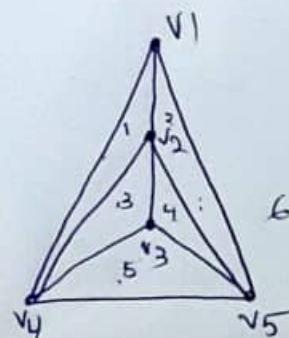
$$e = 8$$

$$f = 8 - 6 + 2$$

$$= 2 + 2 = 4$$

$$= 4$$

$$\begin{aligned} 3 \cdot 2 + 4 + 6 &= 2e \\ 16 &= 2e \\ \underline{\underline{16}} & \end{aligned}$$



$$n = 5$$

$$e = 9$$

$$f = 9 - 5 + 2$$

$$= 4 + 2 = 6$$

Corollary:

In any simple connected planar graph with f regions, n vertices & e edges ($e \geq 2$) the following inequalities must hold.

$$1) e \geq \frac{3}{2}f \quad 2) e \leq 3n - 6$$

Proof: if i have a planar graph.
 *1 each region is bounded by atleast 3 edges.

*2 each edge belongs to 2 Regions

$$2e \geq 3f$$

$$e \geq \frac{3f}{2} \Rightarrow \frac{2e}{3} \geq f$$

we have

$$f = e - n + 2$$

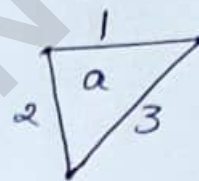
$$f \leq \frac{2e}{3}$$

$$e \geq \frac{3f}{2}$$

$$\frac{2e}{3} \geq e - n + 2$$

$$2e \geq 3e - 3n + 6$$

$$3n - 6 \geq e \quad \text{or} \quad e \leq 3n - 6$$



$$3 \cdot 2 + 4 = 2e$$

$$6 + 4 = 10$$

$$10 = 10$$

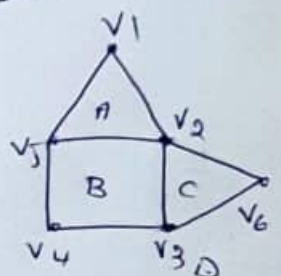
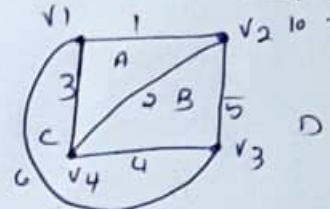


Fig: K_5 $n=5$ $e=10$

$$3n - 6 = 3 \times 5 - 6 = 15 - 6 = 9$$

$$e \leq 3n - 6$$

$10 \leq 9$ not true so K_5 is non planar.

K_{33}

$$n=6 \quad e=9$$

$$f = e - n + 2$$

$$f = 9 - 6 + 2 = 11 - 6 = 5$$

$$e - 2e \geq -2n + 4$$

$$-e \leq -2n + 4$$

$$e \leq 2n - 4$$

There are no triangles in K_{33}

$$2e \geq 4f$$

$$e \geq 2f$$

$$e \geq 2(e - n + 2)$$

$$e \geq 2e - 2n + 4$$

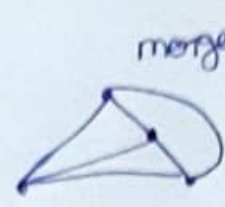
Homeomorphic graphs: or Homeomorphic graph

Two graphs are said to be homeomorphic if one graph can be obtained from the other by the creation of edges in series or by merging of edges in series.

Eg:



(G1)



merger of edges in series



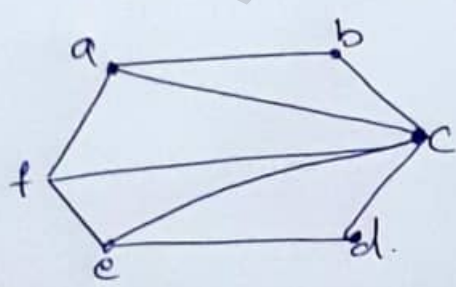
creating edges in series

Homeomorphic graphs

Necessary and Essential condition for planar graph

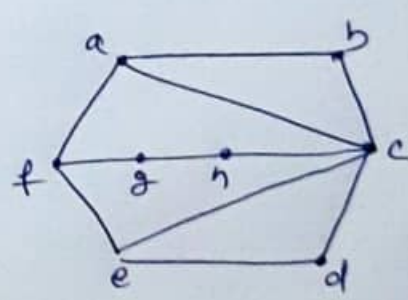
*1) Graph should not contain any of the 2 Kuratowski's graph and any homeomorphic graphs to the Kuratowski's graphs.

Eg: Homeomorphic graphs

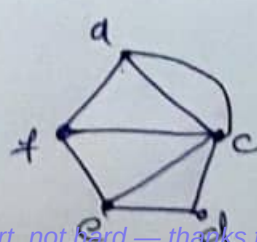
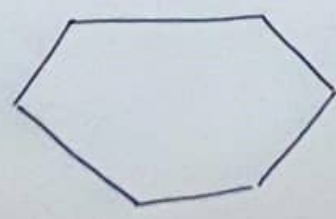


(G1)

*1) Create a new edge it automatically creates a vertex



homeomorphic by creating edge in series



merger of edges

Plane Representation and connectivity;

Disconnected graph:-

* In a disconnected graph the embedding of each component can be considered independently.

* $\therefore$ disconnected graph is planar if and only if each of its components is planar.

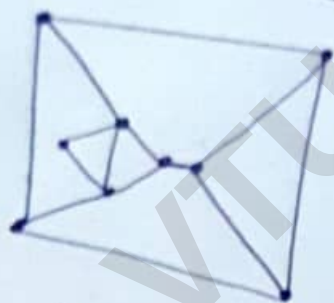
ii) Separable graph

* In Separable graph the embedding of each block can be considered independently.

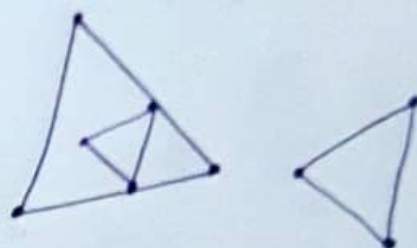
* Hence a separable graph is planar if and only if its blocks is planar.

Disconnected graph:-

Eg:



(a)
connected graph



b
disconnected graph
planar graph



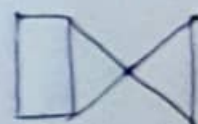
Dis connected graph
non planar



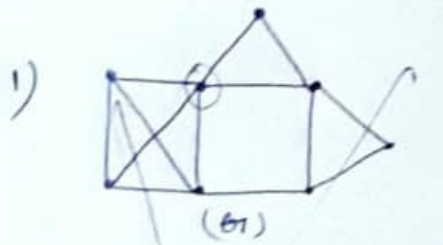
Disconnected
non planar



Disconnected graph
planar.



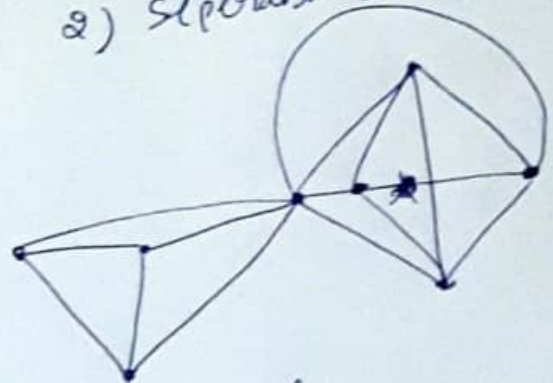
Example:-



separable graph
Planar.



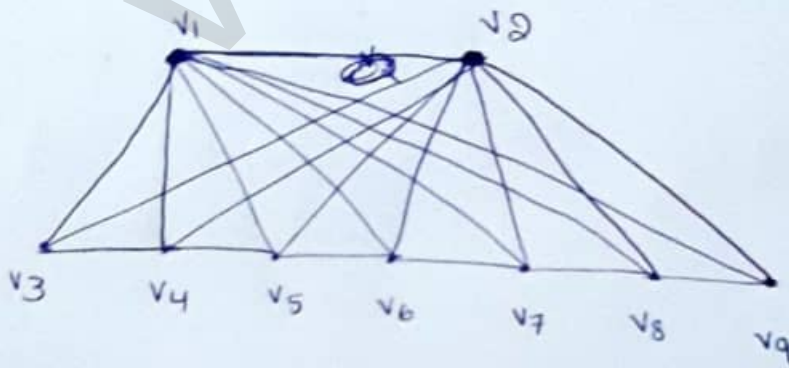
2) Separable graph



separable graph
non-planar.



Embedding graph:-



non-planar



2. connected graph

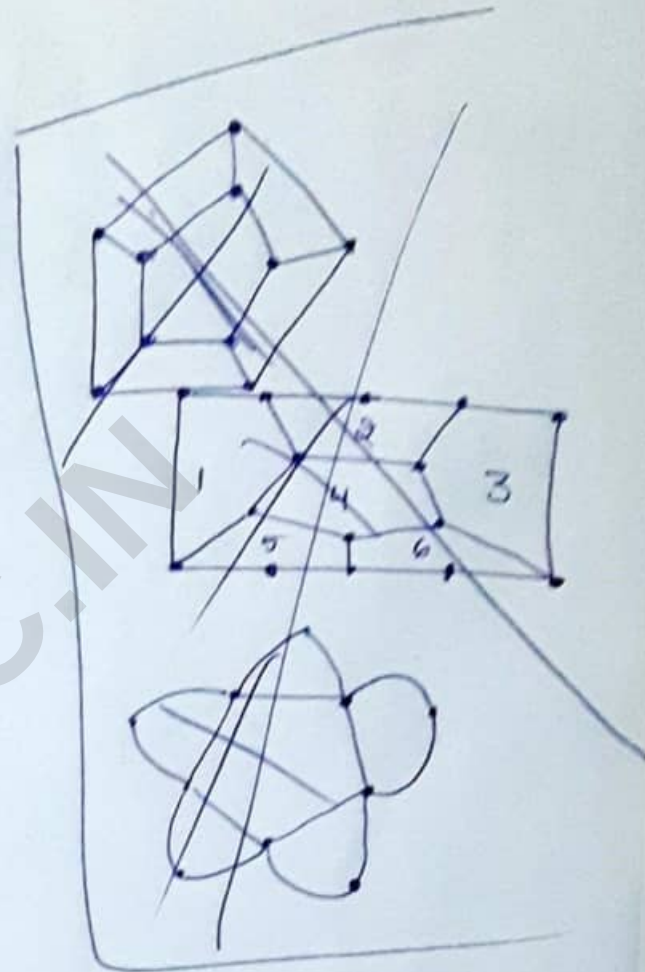
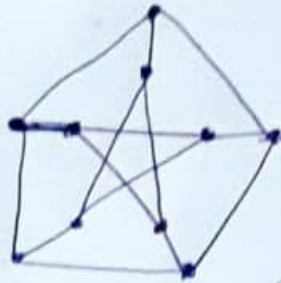


all circuits or loops are a connected graph

3. connected graph:-

Petersen graph

non-planar.



~~3. connected graph~~

Seperable graph:-

a connected graph is said to be Seperable if its vertex connectivity is one.

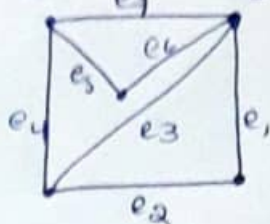
(if you remove one vertex then it will be dis connected graph)

3-connected graph:

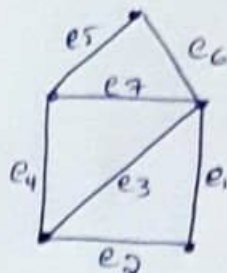
for every pair of distinct vertices $u \in V$ in G
There are atleast 3 internally disjoint paths b/w u

Distinct Embedding:

1)



(a)

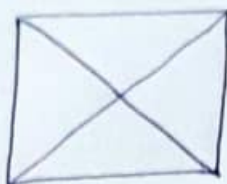


infinite region has 5 edges

infinite region has 4 edges

2 distinct plane representations of graph

2)



2 distinct plane representations

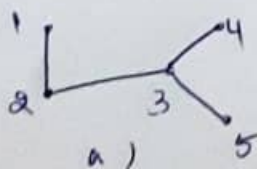
Unique Embedding:-

If of all possible embeddings on a sphere no two are distinct, the graph is said to have unique embedding.

Theorem 7:

The spherical embedding of every planar 3-connected graph is unique.

1-Connected graph:-



a)



not a

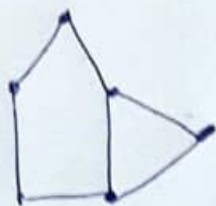
There exactly only (edge disjoint) one path b/w every pair of v

Separable & planar graph examples:-

Numerical on inequalities:

The maximum no of faces that are possible for a simple connected planar graph. n vertices are of degree $(f) \geq 3$.

Ans degree of f means edges of Region.



$$3 + 5 + 6 = 2e$$

$$14 = 2 \times 7$$

$$14 = 14$$

$$2e \geq 3f$$

$$e \geq \frac{3f}{2}$$

$$n - e + f = 2$$

$$n + f = 2 + e$$

$$n + f \geq 2 + \frac{3f}{2}$$

$$n - 2 \geq \frac{3f}{2} - f$$

$$2n - 4 \geq \frac{3f - 2f}{2}$$

$$2n - 4 \geq f$$

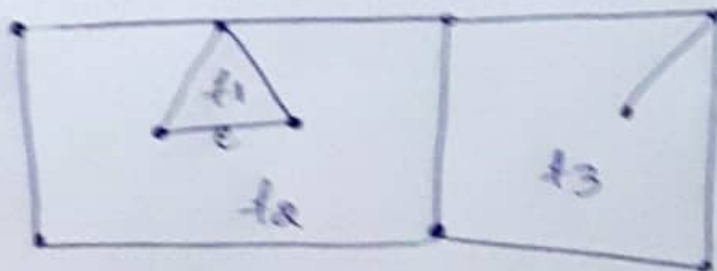
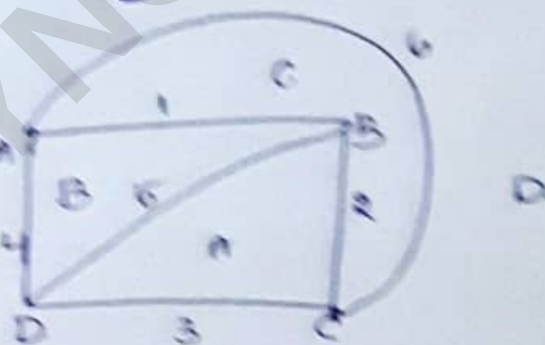
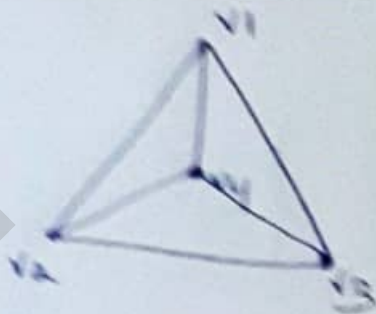
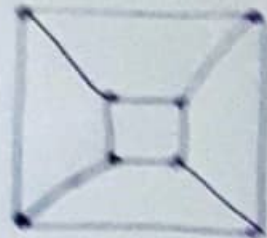
$$f \leq 2n - 4$$

$$f \leq 24 - 4$$

$$f \leq 20$$

Planar graphs:

eg:



14

Matrix Representation:-

Incident Matrix:-

* Let G be a graph with n vertices, e edges and no self-loops.

* we define matrix $A = [a_{ij}]$ of the order $n \times e$ (n by e)

↓ ↓
row column

* n rows corresponds to the n vertices

* e columns corresponds to e edges

* The matrix elements are as below.

$$a_{ij} = \begin{cases} 1 & \text{if } j^{\text{th}} \text{ edge } e_j \text{ is incident on } i^{\text{th}} \text{ vertex } v_i \\ 0 & \text{otherwise.} \end{cases}$$

* $[a_{ij}]$ is known as vertex-edge incident matrix or incident matrix. it is also called binary matrix.

* observations of incident matrix A .

1. Since every edge is incident on exactly 2 vertices each column of A has exactly two 1's.

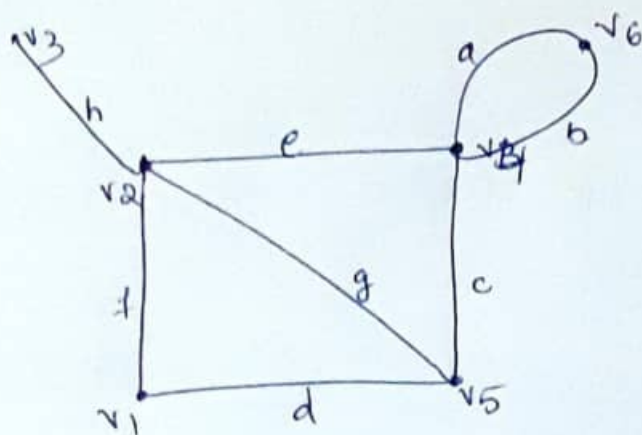
2. Number of one's in each row equals to degree of the corresponding vertex.

3. a row with all 0's represents an isolated vertex.

4. Parallel edges in a graph produces identical columns in its incident matrix.

5. If G is disconnected & consists of 2 components G_1 & G_2 the incident matrix of $A(G)$ of G can be written in block-diagonal form $A(G) = \begin{bmatrix} A(G_1) & 0 \\ 0 & A(G_2) \end{bmatrix}$

6. Permutations of any 2 rows or columns in an matrix simply corresponds to relabeling the vertices & edges of same graph.



(G)

	a	b	c	d	e	f	g	h
v1	0	0	0	1	0	1	0	0
v2	0	0	0	0	1	1	1	1
v3	0	0	0	0	0	0	0	1
v4	1	1	1	0	1	0	0	0
v5	0	0	1	1	0	0	1	0
v6	1	1	0	0	0	0	0	0

6x8

Theorem - 2

*1 If $A(G)$ is an incidence matrix of a connected graph G with n vertices, the rank of $A(G)$ is $(n-1)$.

Corollary of 2 Theorem - 3

The reduced incidence matrix of a tree is non singular.

Theorem 4

Permutation (& combination) of any 2 rows or columns in an incidence matrix simply corresponds to relabeling the vertices and edges of the same graph.

Permutation is - rearrangement of existing elements.

*1 Two graphs G_1 & G_2 are isomorphic iff their incidence matrices $A(G_1)$ & $A(G_2)$ differ only by permutations of rows & columns.

$$nPr = \frac{n!}{(n-r)!}$$

Circuit Matrix:

- 1) Let no of different circuits in a graph G be a & no of edges in G be e .

then circuit matrix $B = [b_{ij}]$ of G is a $a \times e$ matrix defined as

$$b_{ij} = \begin{cases} 1 & \text{if } i^{\text{th}} \text{ circuit includes } j^{\text{th}} \text{ edge} \\ 0 & \text{otherwise.} \end{cases}$$

$$B(G) = \begin{matrix} & \begin{matrix} a & b & c & d & e & f & g & h \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

Circuit Matrix observation:-

1. A column of all zeros corresponds to a non circuit edge (edge that ^{does not} belong to any circuit)
2. Each row of $B(G)$ is a circuit vector.
3. Unlike incident matrix, a circuit matrix is capable of representing a self-loop - the corresponding row will have a single 1.

4. Number of 1's in a row is equal to the Number of edges in the corresponding ckt.

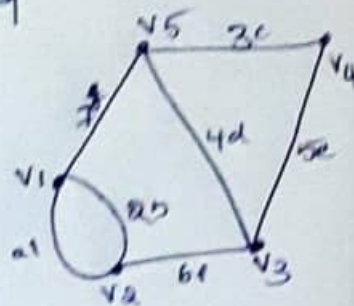
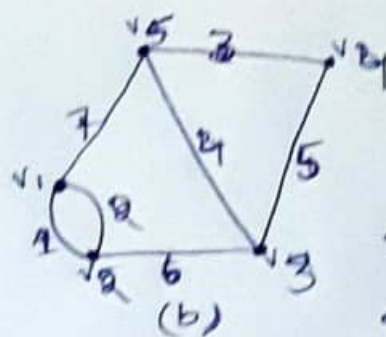
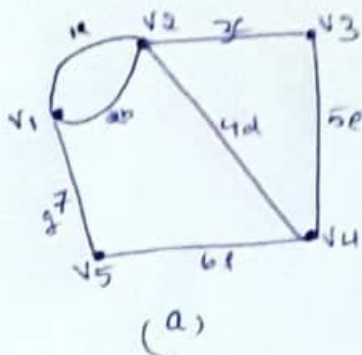
5. If graph G is separable (or disconnected) and consists of two blocks (or components) G_1 and G_2 the circuit matrix written in block diagonal.

$$B(G) = \begin{bmatrix} B(G_1) & 0 \\ 0 & B(G_2) \end{bmatrix}$$

6. permutation of any two rows or columns in ckt matrix simply corresponds to relabeling the circuits and edges.

7. circuit matrix does not specify a graph completely.

Ex: of circuit matrix



- circuit 1 $\rightarrow \{1, 2\}$
 circuit 2 $\rightarrow \{3, 4, 5\}$
 circuit 3 $\rightarrow \{2, 4, 6, 7\}$
 circuit 4 $\rightarrow \{3, 5, 6, 7, 2\}$
 circuit 5 $\rightarrow \{3, 5, 6, 7, 1\}$

- circuit 1 $\rightarrow \{1, 2\}$
 circuit 2 $\rightarrow \{3, 4, 5\}$
 circuit 3 $\rightarrow \{2, 4, 6, 7\}$
 circuit 4 $\rightarrow \{3, 5, 6, 7, 2\}$
 circuit 5 $\rightarrow \{3, 5, 6, 7, 1\}$

$$B[b_a] = \begin{matrix} & \begin{matrix} a & b & c & d & e & f & g \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

$$B[b_b] = \begin{matrix} & \begin{matrix} a & b & c & d & e & f & g \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

But graph (a) & (b) are not isomorphic.

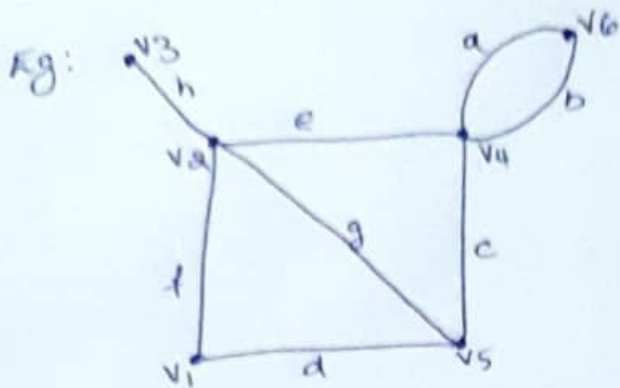
Theorem-4:

Let B and A be respectively circuit matrix and the incidence matrix (of a self-loop-free graph) whose columns are arranged using same order of edges.

Then every row of B is orthogonal to every row A

$$\therefore A \cdot B^T = B \cdot A^T = 0$$

where T represents transposed matrix



$$A(G) = \begin{matrix} & \begin{matrix} a & b & c & d & e & f & g & h \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$B(G) = \begin{matrix} \begin{matrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{matrix} & \begin{matrix} a & b & c & d & e & f & g & h \end{matrix} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$B^T(G) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A \cdot B^T = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

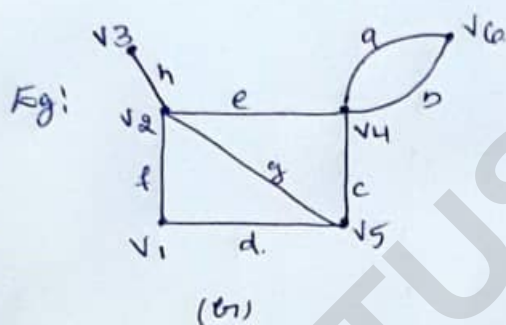
Path Matrix:

- * Another (0,1)-matrix this matrix is convenient to use in communication and transportation networks is the path matrix.
- * path matrix is defined for a specific pair of vertices in a graph say (x, y) & is written as $P(x, y)$
- * rows corresponds to different path b/n vertex x & y
- * columns corresponds to the edges in E .

$$P(x, y) = [P_{ij}]$$

$$P_{ij} = 1 \quad \text{if } j^{\text{th}} \text{ edge lies in } i^{\text{th}} \text{ path.}$$

$$= 0 \quad \text{otherwise.}$$



Consider the paths b/n vertices v_3 & v_4 of given graph 'b'.

* There are 3 paths b/n v_3 & v_4

Path 1 $\rightarrow \{h, e\}$

Path 2 $\rightarrow \{h, g, c\}$

Path 3 $\rightarrow \{h, f, d, c\}$

We get 3×8 path matrix $P(v_3, v_4)$

$$P(v_3, v_4) = \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \begin{bmatrix} a & b & c & d & e & f & g & h \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Observations of Path matrix

1. a column of all 0's corresponds to an edge that does not lie in any path b/n x & y
2. A column of all 1's corresponds to an edge that lies in every path b/n x & y .
3. There is no row with all 0's
4. Sum of any two rows in $P(x, y)$ corresponds to a circuit or an edge disjoint union of cuts.

Eg:

	a	b	c	d	e	f	g	h
Path 1	0	0	0	0	1	0	0	1
Path 2	0	0	1	0	0	0	1	1
Path 3	0	0	1	0	1	0	1	0

Path 2 + Path 3 $\Rightarrow$ circuit

	a	b	c	d	e	f	g	h
Path 2	0	0	1	0	0	0	1	1
Path 3	0	0	1	1	0	1	0	1
Path 3 + Path 1	0	0	0	1	0	1	1	0

Path 3 + Path 1 $\Rightarrow$ cut

	a	b	c	d	e	f	g	h
Path 3	0	0	1	1	0	1	0	1
Path 1	0	0	0	0	1	0	0	1
Path 3 + Path 1	0	0	1	1	1	1	0	0

Path 3 + Path 1 $\Rightarrow$ cut

Theorem 5: If A is the incident matrix & $P(x, y)$ is the path matrix of vertex x & y belongs to A and columns of both the matrix have same order for edges then the product $A \cdot P^T(x, y) = M$, where M has 1's in two rows x & y and rest of $(n-2)$ rows are zeros.

Proof:

$$A = \begin{matrix} & \begin{matrix} a & b & c & d & e & f & g & h \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}_{6 \times 8}$$

$$P^T(x, y) = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \\ g \\ h \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \end{matrix}_{8 \times 3}$$

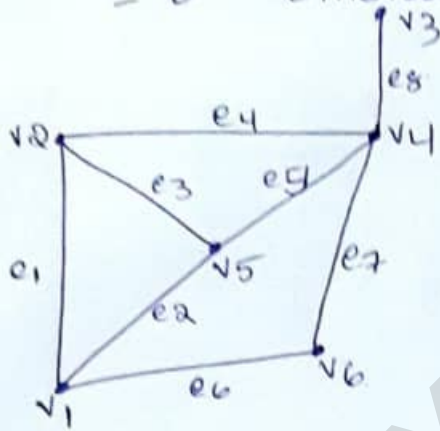
$$A \cdot P^T(x, y) = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}_{6 \times 3}$$

Adjacency Matrix:

- 1) alternative to incident matrix is adjacency matrix.
- 2) it is convenient to represent a graph by adjacency matrix or connection matrix. represented by $X = [x_{ij}]$
- 3) Adjacency matrix of a graph G with n vertices & no parallel edges is $n \times n$ symmetric binary matrix

$$x_{ij} = \begin{cases} 1 & \text{if there is an edge b/w } i^{\text{th}} \& j^{\text{th}} \text{ vertex} \\ 0 & \text{Otherwise} \end{cases}$$

Eg:



$$X = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

Observations of Adjacency Matrix:-

1. The entries along principle diagonal of X are all 0's if graph has no self loop else $x_{ii} = 1$ for i th vertex self loop
2. no parallel edge is represented by adjacency matrix
3. If graph has no self-loops, the degree of vertex equals the no of 1's in corresponding row or column of X .
4. Permutations of rows and corresponding columns imply reordering the vertices

$$X(G_2) = R^T \cdot X(G_1) \cdot R,$$

R is permutation matrix

5. A graph G_1 is disconnected & has 2 components G_1 & G_2 then $X(G_1)$ is

$$X(G_1) = \begin{bmatrix} X(G_1) & 0 \\ 0 & X(G_2) \end{bmatrix}$$

6. Given any square, symmetric binary matrix A of order n , one can always construct a graph of n vertices such that A is the adjacency matrix

$$X \cdot X = X^2$$

$$X^2 \cdot X = X \cdot X^2 = X^3$$